College of Engineering

Department of Mechanical Engineering

ME 340 - Introduction to Manufacturing Engineering

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HAAS CNC Mill Report

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Problem Definition

- While employing the proper safety procedures, load the proper drill bits into the HAAS CNC mill, establish the home positions, load the tool route program, and run the machine to produce a finished part.

Deliverables

- A finished 3x3 metal square with the given design.
- A detailed report on the experiences and actions of the operation.

Procedure

1. Obtain the 0.250 4-flute End Mill, 0.250 Drill, 0.250 Spot Drill, 0.250 4-flute Ball End Mill, and Electronic Edge finder.
2. Load the program from machine memory by pressing LIST PROG, view the MEMORY tab, use the arrows to highlight the program, press WRITE/ENTER, enter Program O34002, then press SELECT PROGRAM. Double check the program comments to ensure tool numbers match your expectations.
3. To load a tool into the milling machine, first, close the doors, press MDI, type T1, press ATC FWD, open the doors, and insert the 0.250 4-flute End Mill by lining up the key with the spindle slot and pressing the black button.
4. Repeat step 3 for the 0.250 Drill, 0.250 Spot Drill, 0.250 4-flute Ball End Mill, and Electronic Edge finder for slots 2, 3, 4, and 20, respectively.
5. To clamp the part into the vice, press HAND JOG, bring the vice closer using the xyz buttons, place the raw part into the jaws, and close the jaws with the vise handle until snug.
6. Ensure the workpiece extends at least 0.375” above the vice using calipers.
7. Hammer the top of the part once with the orange deadblow hammer or until the part is snug with no gaps between the vice.
8. To set the home on the back left corner of the part, in HAND JOG mode lower the edge finder beside the left edge of the workpiece until the light turns on using “0.01” for quick moves and “0.001” for small moves. On the control panel, press OFFSET, highlight the x field on line G54, press PART ZERO SET. In HAND JOG mode, lift the edge finder above the part and position the x-value to 0.1000 to account for the radius of the edge finder tip. Return to OFFSET, line G54, and press PART ZERO SET again to re-zero the x-axis.
9. Repeat step 8 for the y-axis on the back of the workpiece.

10. To set up tool length offset, call a tool to the spindle by pressing MDI, typing T#, and pressing ATC FWD. In HAND JOG, position the tool 0.125” above the center of the part. Place a piece of paper over the part and lower the tip of the tool in 0.001” increments until the paper just starts to feel friction when moved back and forth. Press OFFSET to activate the TOOL OFFSET table, and then press TOOL OFFSET MEASURE. Call the next tool by pressing MDI, typing T#, and pressing ATC FWD.
11. Repeat step 10 for all four tools.
12. To check coolant, call tool 1, close the doors, press COOLANT, and ensure the stream is flowing and directed toward the tool tip.
13. To run the program, press LIST PROG, select MEM, press 25% RAPID, press SINGLE BLOCK, and select CYCLE START. Traverse the code one line at a time to ensure the right commands. Stop the program when the tool is 3” above the part and use the 123 block to confirm the height is correct. Repeat this for each tool. When confirmed, press CYCLE START. Keep hand near EMERGENCY STOP in the case of anomalies.
14. When finished, open the doors, air gun the chips and coolant off of the piece, open the vice jaws, and remove the final part.
15. Remove and return all of the tools.

Reflection

To set up the HAAS CNC machine requires attention to detail in order to follow the instructions, traverse the computer interface, load the tools, set their home positions without breaking them, and ensuring the mill follows the correct tool path. Overall, the machine does not require any particular skill set, but if instructions are not followed very precisely, the drill heads could shatter, the mill could break, the part could be ruined, or someone could get seriously injured.

Work



Figure 1: HAAS VF2YT 3-Axis CNC Vertical Milling Center

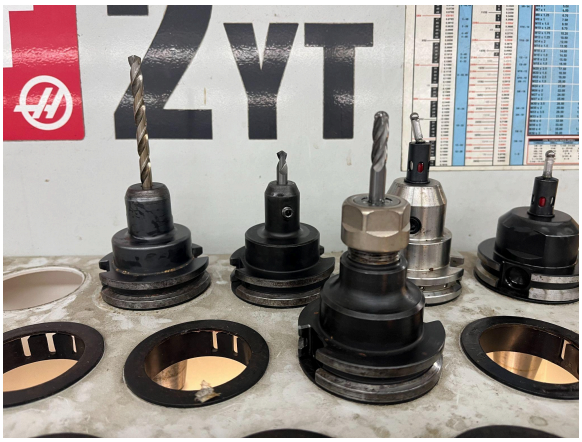


Figure 2: Here are some of the drill bits used from carving the part as well as the edge finders used to home the tools.



Figure 3: The main stage in the CNC mill including the vice, tool slot, and automatic tool changer (ATC).

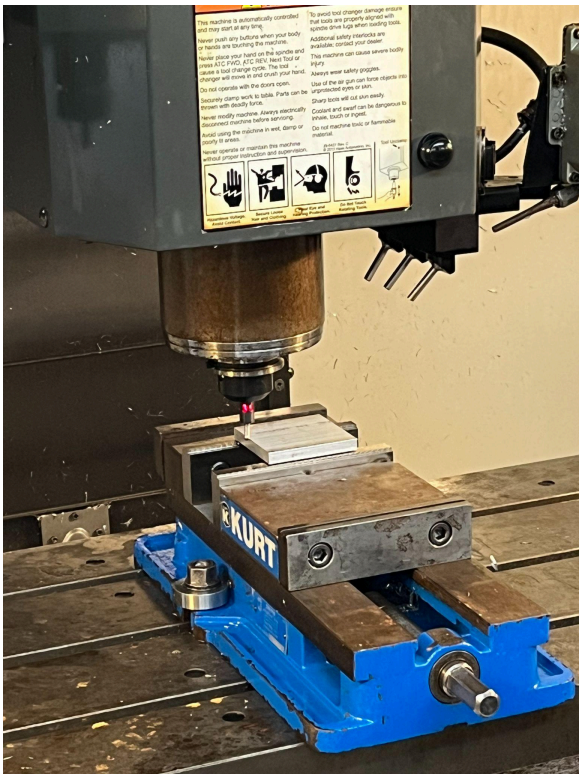


Figure 4: The edge finder glows red when it first pushes on the edge of the part.



Figure 5: The computer interface to manage the machine functions and set parameters.



Figure 6: The 123 block on its 3” side to set the drill height.



Figure 7: While the drill operates, a thorough supply of coolant is sprayed to prevent overheating and clear debris.



Figure 8: The 3x3 metal square before milling.

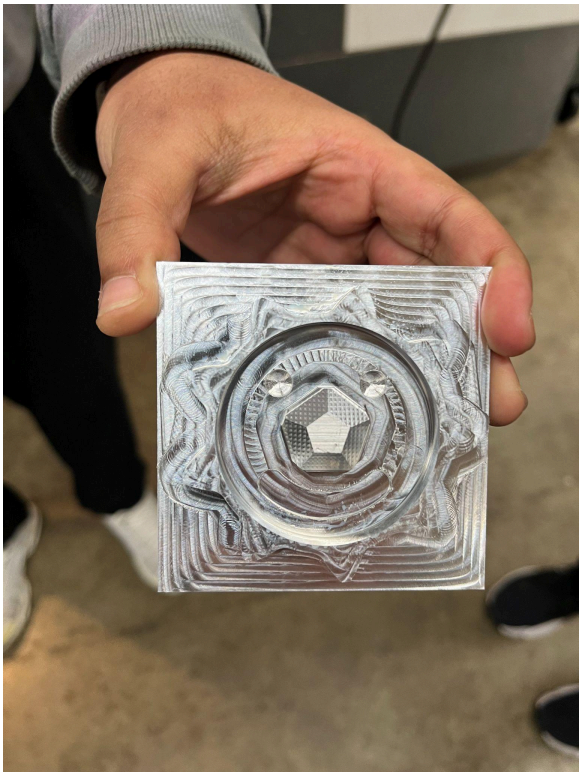


Figure 9: The finished product after milling.

Conclusion and Recommendations

What parts of the lab went well?

- Almost everything in the lab went smoothly and efficiently from loading the tools and program to setting the offsets and running the program.

What did you struggle with?

- The most difficult parts of the lab was the repetitiveness and precision in setting the xyz offsets. Passing the tool a couple thousands over the zero marks could shatter the drill bits, so these steps required extra caution. Then, repeating these steps for all four tools while maintaining the same level of caution became very laborious.

Did you find any part of the lab to be unclear or repetitive?

- No part of the lab was unclear. However, the process of homing each tool became very repetitive when moving the tool down thousands of an inch at a time to touch the paper needed to be repeated four times.

If you were tasked to make 100 of these parts, what could you change about the process to speed up production?

- In order to mass produce these parts, it would be wise to find a way around setting the offsets for each piece when swapped out. Hence, it would be much more practical to cut

out multiple pieces from one larger sheet of metal. This would tremendously reduce load time and homing time.

How would you machine features on the part that aligned with the position of features on the back of the part? Would anything about the setup you did today need to change?

- The best way to align features on the front and back would simply be through equal measurements. However, since the piece is flipped, the measurements from the right edge on the top would need to be made from the left edge on the bottom.